

Reviewer Notes:

Reviewer #1: Kang et al. propose a modelling study to inform benefit and risks of recommending Ixchiqu vaccine in diverse exposure situation. It is a welcome study in the context of limited data, epidemiological uncertainty and changing recommendations. While the conclusions are interesting, the author should be more transparent about the model assumptions and parameters, as well as in the presentation of the results.

Introduction

A description of the Brazilian surveillance system and Chik epidemiology is welcomed

Methods

1) Model specification : what is population modeled, is there a time horizon? Authors should explicitly say (if I understood correctly) it is a deterministic ordinary differential equation model/compartamental model/SEIR based model.

2) There should be a paragraph on model calibration in the main manuscript

3) Disease progression parameters: Please add the references or mention briefly the source for DW

4) Authors should include a table with key model parameters in the main manuscript

5) SAE Input data

- I suggest to update the incidence of SAE from latest publication : PMID: 4186391

- For more accuracy, the authors could also use the prevalence of anti Interferon antibodies, which could explain SAE in patients. <https://doi.org/10.1073/pnas.25322121>

6) Has BRR been used in previous modeling studies? authors should provide guidance on how interpret the results

7) The use of the word "state" is ambiguous. Could refer to the Brazilian States or the Health states of the model.

8) Not clear the source used to estimate baseline immunity

9) PSA: not clear how many second order simulation the authors performed : 10,000 simulations of the initial population of 10,000 individuals?

10) Who is the funding source?

11) If I understand correctly, outbreak response immunization, with protection against disease and infection, has a herd effect? in this scenario, do vaccinations occur at the beginning of the model?

RESULTS

11) There should be a paragraph on model validation

12) Presentations results : authors only provide the BRR, whilst in the methods they claim they would present SAEs, Death and DALYs

13) travel vaccination > low transmission : I do not understand why authors said the risks outweigh benefits, while their BRR > 1, 91.5% probabilities of benefits outweighing risks (line 315). Is the probability given by PSA?

Graph could be more easy to read than tables

DISCUSSION

14) I think that BRR for long travels (90 days) should match BRR in outbreaks? Has it been tested in sensitivity analysis

15) It should be interesting to discuss the BRR for travelling if it occurs during an outbreak or not

16) It would be nice to confront these results with other studies using BRR

17) How would BRR change with different baseline serorevalence?

Reviewer #2: Summary

I would like to thank the authors for tackling an important and genuinely timely question. The balance of benefits and risks for Ixchiqu, considering the post licensure safety signals and the remaining uncertainty about its mechanism of protection, is a pressing policy issue, and the effort to build a structured, quantitative framework around it is valuable. Having said that, I have concerns about the modelling framework that, in my view, affect the reliability of the paper's central conclusions, as well as concerns about how the results are presented for the broad readership of PLOS Medicine. On balance, I do not think the manuscript is ready for publication in its current form, and I would recommend rejection, while hoping the comments below are useful should the authors wish to pursue this work further.

Major comments

1) On the identifiability of the transmission model, and how it interacts with the characterisation of prior immunity.

I looked closely at how the model separates two things that, in a compartmental framework, are fundamentally hard to tell apart. A decline in weekly case counts can be produced either by depletion of the pool of susceptible people, meaning enough people have already been infected that the outbreak runs out of people to infect, or by a genuine drop in the transmission rate itself, caused by behaviour change, vector seasonality, or control measures. **The model fits 52 independent weekly transmission rate parameters per state, each treated as a free parameter with only a loosely informative prior and no smoothing or correlation linking one week to the next.** A model with this much freedom can reproduce almost any observed case trajectory, regardless of which of the two explanations above is actually true.

This matters a great deal here because the initial pool of susceptible people, S_0 , is not estimated jointly with the transmission rate. It is fixed in advance, using a simple catalytic hazard model. This age specific proportion is averaged across each age band, multiplied by the population in that band, and summed across all ages to give S_0 for the state, all before the weekly transmission parameters are fitted to the 2022 case data. Because those weekly parameters are so flexible, the fitting exercise can converge on a convincing fit to the observed case curve almost regardless of whether this fixed value of S_0 is correct. If S_0 is too high, the fitted transmission rate can simply decline faster to compensate, and the reverse is equally possible. So the fit shown in the supplementary figures, while visually convincing, is not on its own evidence that the baseline immunity assumption is right. It may just as easily reflect the model absorbing that error into the transmission

rate estimates instead. **I would ask the authors to demonstrate that this is not happening in practice, for example through a parameter recovery exercise on simulated data with a known true value of S_0 , or by refitting the model under two or three alternative baseline immunity assumptions and showing that the downstream vaccination results are not sensitive to that choice.**

I would also note that the wording in the supplement adds to this ambiguity. Section S1 describes S_0 as output from the fitted transmission model evaluated at the initial time point, which is not quite accurate. **The value at the initial time point is the starting condition supplied to the model before fitting begins, not something the fitting procedure itself estimates.** I would ask the authors to state this distinction clearly, since a reader could otherwise reasonably, but incorrectly, assume that S_0 is being estimated together with the transmission rate rather than fixed beforehand.

This is compounded further by a **spatial concern that the authors raise themselves but do not fully reconcile.** In their limitations, they note that local serosurveys point to substantial variation within a state in underlying immunity. **If the observed case counts in a given state actually reflect several distinct, spatially separated outbreaks occurring at different times, which is plausible and well documented for a vector borne disease such as chikungunya, then**

fitting one uniform epidemic curve with a single time varying transmission rate to that combined case series risks producing a fitted trajectory that is really an artefact of several overlapping local outbreaks rising and falling at different times and in different places, rather than a genuine, interpretable transmission parameter for the state as a whole. I think the authors need to explain how this acknowledged spatial heterogeneity is reconciled with the decision to fit a single combined curve per state, since as it stands the two appear to be in tension with one another.

I want to stress why this particular chain of assumptions matters more than a routine calibration caveat. It is precisely the size of the remaining pool of susceptible people, the very quantity whose estimation is compromised by the identifiability problem described above, that determines the size of the indirect, herd protection generated once vaccination is assumed to block infection rather than disease alone. Under the assumption that the vaccine blocks both infection and disease, the benefit attributed to vaccination depends directly on how many people are removed from an already uncertain pool of susceptibles, and on how that changes onward transmission. If the true number of susceptible people at the time of vaccination is smaller than assumed, because more people are already immune than the catalytic model implies, or because the fitted transmission rate has quietly compensated for an incorrect starting value, the model would overstate the indirect benefit of vaccination that blocks infection. If the true number is larger, it would understate it. This is exactly the mechanism responsible for the paper's central finding that older adults only benefit when the vaccine is assumed to block infection, which makes it, in my view, the single point in the analysis most in need of a robustness check before that conclusion can be relied upon.

2) On the assumption that the vaccine blocks infection.

Building on the point above, the model only ever compares two fixed extremes, namely blocking disease only versus blocking both disease and infection, applying the same efficacy to both, without exploring any intermediate degree of protection. Given how wide the reported uncertainty intervals already are, I suspect that formally including uncertainty around this mechanistic assumption, rather than treating it as a simple switch between two states, would widen those intervals further still. I think this would strengthen the honesty of the paper about what is and is not currently known, and I would encourage the authors to consider it.

3) On the figures.

The figures in this manuscript are difficult to interpret and, in their current form, function as model diagnostics rather than as figures suited to a main text article. Figure 1B presents twelve panels on a logarithmic scale, each showing tens of thousands of overlapping simulated draws, split by age group and by travel duration and transmission setting. This same pattern repeats across seven related panels for Figure 1 and seven more for Figure 2, meaning the reader must work through fourteen dense, multi panel plots to extract the paper's central message. Figures of this kind are appropriate for a modeller checking the full output of a probabilistic sensitivity analysis, but they are not appropriate as the primary visual communication of a clinical and policy paper aimed at the broad readership of PLOS Medicine. I do not think this is a matter of style. As presented, these figures do not convey the paper's findings, they display them, and the two are not the same thing. I

would ask the authors to replace them in the main text with a small number of solid simplified summary figures, of the point and interval type standard in benefit risk reporting, and to move the full probabilistic output, including Figure 1 and Figure 2 as currently constructed, to the supplement, where it belongs alongside the correlation and parameter distribution figures already placed there.

4) On the tables.

The tables share the same problem and need to be substantially shortened before this paper is suitable for publication. Table 1, by my count, runs to roughly seventy rows of results across every combination of outcome, transmission setting, age group, and travel duration, each with

an accompanying uncertainty interval. Table 2 is comparably long. Tables of this length are not readable as main text tables. They read as a full data export rather than a table designed to make a point, and I do not think a reader of this journal can be expected to locate the paper's key findings within them. I would ask the authors to reduce the main text tables to a single, focused table, for instance restricted to the DALY outcome, with the full breakdown by outcome, setting, age group, and duration moved to the supplement in its entirety.

5) On the travel vaccination estimates.

I would like to acknowledge that the travel risk calculation thoughtfully uses the fitted, time varying daily force of infection from the 2022 outbreak rather than a fixed long term average, and samples the day of entry to capture some variation in timing within that outbreak. **That said, it still relies on a single average for the whole state, applied to one specific historical outbreak season, and it inherits the same concerns about spatial aggregation raised in point one.** I would encourage the **authors to discuss how the well documented spatial focality of chikungunya transmission, together with year to year variation in outbreak size and timing, might affect how representative these estimates are of the exposure a traveller would actually face.**

6) On the novelty of the contribution.

I want to be careful here, since the authors have put together a detailed framework than what currently exists in the literature. That said, I think it is worth situating the contribution more precisely relative to two closely related pieces of work: Cracknell Daniels and colleagues, published in Nature Health in January 2026, who proposed a generalised framework for assessing benefit and risk for vaccines approved through accelerated pathways, including Ixchiq, and the authors' own earlier paper, Kang and colleagues, published in EClinicalMedicine in December 2025, from which the criteria for selecting states and the approach to baseline immunity used here both appear to be drawn. **The genuinely new elements in this submission, namely the distinction between blocking disease alone and blocking both infection and disease, and the serostatus adjusted risk scenario, are, as discussed in point one, exactly the components resting on the least certain parts of the model. I would encourage the authors to state more explicitly what is new here once that overlap is taken into account.**

Recommendation

Reject, though I hope constructively so. My central concern is that the paper's most policy relevant conclusion, that outbreak response vaccination benefits older adults only when the vaccine is assumed to block infection, rests on a pool of susceptible people that is fixed from an unvalidated and spatially coarse immunity assumption, fitted using a transmission rate model flexible enough that it cannot currently demonstrate this assumption is correct, and layered on top of an acknowledged but unreconciled tension between modelling each state as a single population and the documented spatial heterogeneity of transmission within it. Because it is precisely this pool of susceptible people that drives the indirect, herd protection effects underlying the infection blocking scenario, this is not a peripheral calibration issue but one that sits at the heart of the paper's main conclusion.

Reviewer #3: General:

Important topic as the threat of chik grows and there are two vaccines which have different characteristics. The issue is identifying the appropriate data to assess safety and effectiveness. Extrapolation from passive animal transfer and natural infection studies is complex. Using VAERS is a major limitation due to the unverified nature of the data. Using the Reunion Island data is also an issue based on who was preferentially vaccinated and whether or not it was appropriate considering the warning which existed for the vaccine based on phase 3 safety data (more severe adverse events in older adults). Factors impacting risk-benefit are complex and

need to be very clearly teased out and how the limitations in data affect the overall limitations of the computations. There are also some important but missing references and data which have been added at the end of the report.

Abstract:

Introduction:

"virus-like particle vaccine (Vimkunya, ≥ 12 years)."

- Where is it licensed or authorized?

"restricted ability to detect rare adverse events"

- Would improve the balance of the manuscript to describe why this is the required path for vaccines against diseases like chik and the post-licensure requirement studies which sponsors must complete to maintain their license.

"mass vaccination campaign "

- Important to describe this campaign in more detail. The government ran the campaign and preferentially vaccinated older adults.

"(VAERS)"

- A statement about the limitation of this database is required.

Methods:

"probabilities from the US VAERS data"

- Again, a limitation these are not verified and anyone can submit a report.

"Among 18,445 vaccinated older"

- Where did this data come from? Reference?

Outcomes:

"testing PCR-positive for the vaccine strain"

- Interesting to know how frequently this information was available.

"we assumed that travellers experienced the same daily infection risk"

- Assume this is based on a region with a predominantly non-immune population due to absence of previous CHIKV transmission?

"protection against disease only, and (ii) protection against both disease and Infection"

- Vaccines are largely developed to reduce the risk of disease. Preventing infection can be a high bar. Why separate them here?

- Are the authors using asymptomatic infection rates in their calculations?

- Are the authors assessing risk of unvaccinated over time as populations become immune from infection and immune from vaccination as campaigns roll out? Risk of vaccination does not change over time but potential benefit would.

Results:

Discussion:

"while chronic sequelae prevented"

- This is a major limitation because the chronic sequelae make up a major component of the public health burden created by CHIKV infections and disease.

"By including chronic sequelae"

- Important to clearly indicate how the incidence of chronic sequelae were estimated considering the variance in the literature.

"informed by US VAERS"

- The biggest issues with VAERS is that reports can be made by anyone and they are not confirmed unless an agency specifically decides to do so for a particular vaccine. This is a

major concern with this manuscript and the data used to inform the model.

These would be a good sources of information not currently referenced:

- Vondeling GT, Croda J, Jelinek T, Kassianos G, Kollaritsch H, Nguyen LBL, Ribeiro GS, Schmidt-Chanasit J, Thomas SJ, Unger Z, de Roo A, Plennevaux E, Jaramillo JC. Post-marketing safety evaluation of the live-attenuated chikungunya vaccine (IXCHIQ). *Vaccine*. 2026 Apr 19;79:128491. doi: 10.1016/j.vaccine.2026.128491. Epub 2026 Mar 20. PMID: 41863914.
- Yoon IK, Alera MT, Lago CB, Tac-An IA, Villa D, Fernandez S, Thaisomboonsuk B, Klungthong C, Levy JW, Velasco JM, Roque VG Jr, Salje H, Macareo LR, Hermann LL, Nisalak A, Srikiatkachorn A. High rate of subclinical chikungunya virus infection and association of neutralizing antibody with protection in a prospective cohort in the Philippines. *PLoS Negl Trop Dis*. 2015 May 7;9(5):e0003764. doi: 10.1371/journal.pntd.0003764. PMID: 25951202; PMCID: PMC4423927.
- Yoon IK, Srikiatkachorn A, Alera MT, Fernandez S, Cummings DAT, Salje H. Pre-existing chikungunya virus neutralizing antibodies correlate with risk of symptomatic infection and subclinical seroconversion in a Philippine cohort. *Int J Infect Dis*. 2020 Jun;95:167-173. doi: 10.1016/j.ijid.2020.03.073. Epub 2020 Apr 1. PMID: 32247051.

Reviewer #4: OVERALL

Many things about this are good. The main things I think need addressing are in summary:

- assumption of identically zero deaths in some cases needs to be changed to a distributional assumption
- fewer figures: I've made suggestions for which I think are most useful and how to regroup
- better organized code & inclusion of data: currently not reproducible that I can see
- improved reporting: self-contained captions; MCMC methods/results; use of appropriate reporting checklist

ABSTRACT

There are quite a few undefined acronyms which impair readability.

SUMMARY

'Moderate' - needs defining here to be self-contained.

Strong overlap between 'findings' and 'meaning' modules.

METHODS

Rationale for attach rate threshold choice? Not my area, but does attack rate have an implicit time horizon/unit, e.g. year?

~ line 250: no point introducing the Greek here as these parameters are not later used in the

main text.

RESULTS

DISCUSSION

I was expecting a return to discuss results in light of ref 22 mentioned in the Introduction, but didn't see it.

TABLES

I think you need to be more explicit with 'Setting' = 'Low' etc, i.e. use a more explanatory column heading and state the definitions in each row-block.

The meaning of 'adjusted' should be explained in the caption of the tables.

'beneficial' similarly should be defined in the table caption. Moreover, the term seems misleading and the calculation confusing. I'm looking at the tables/figures before reading the results, but I don't understand why in a PSA you end up with zero risk of deaths - that doesn't seem a reasonable interpretation of any data for modelling. Zero observed deaths out of N events doesn't mean an identically zero risk, but rather a distribution close to zero. I say the term is misleading because the vaccine is beneficial (expected benefits > 0) in all the cells, so it seems odd to specifically label the cells where expected risk is zero as 'beneficial'. And as I say, expected risk being identically zero can only be due to a modelling choice rather than data.

Abbreviations like DALYs, SAEs should be spelled out in the table captions

FIGURES

Figure 1a - Nice figure. Please spell out abbreviations in the caption and explain the meaning and reason for missing states to make it self-contained.

Figures: BR plane scatter plots - it just ends up with too many plots & I don't think you need these as well as the CEAC plots as they convey much the same information in a harder to read way. Moreover, as currently saved they result in a huge file size (65Mb) due to all the points being individually plotted (rather than hexbinned or density plotted).

With a better aspect ratio, the CEAC plots could make a 3-panel Figure 2 (with Figure 1a being Figure 1) - one panel for each outcome. As with the tables, I don't see how one can justify modelling identically zero deaths.

Suggest Figure 2a becomes Figure 3, & Figure 2e,f,g are stacked to make Figure 4 as with travel, and ditching or appendicizing the scatter plots (with file size reduction if retained somewhere).

CODE & DATA

There are few issues with the repository that inhibit or prevent reproducibility:

1. There is no README/runbook that explains how the analysis pipeline works
2. There are multiple versions of files - brazil_all_draws* has 4 versions, so I am not sure which

is/are used. The 'old_codes' folder should probably be deleted.

3. Given the above, it is hard for me to be sure, but I don't think the data required to run the analysis have been provided. E.g. setup.R (which might reasonably be assumed to be early in the pipeline) loads over 20 Rdata files from the 01_Data/ folder, which contains none. Some of these files may be generated by other parts of the workflow, but I doubt they all stem from the simple single CSV file present in 01_Data/. I didn't spend a lot of time trying to reconstruct how the pipeline works.

SUPPLEMENTARY INFORMATION

Transmission model: I struggled to understand this alongside the code. Is this the stan model? The stan model in wp1_ve_site_priority.R doesn't seem to have explicit structure that maps obviously onto the SEIR framework. Please elaborate.

I didn't see any information on MCMC conditions or convergence reported.

I didn't see any reporting checklist: I would have thought CHEERS 2 might apply given the health economic angle here.

Reviewer #5: The live-attenuated chikungunya vaccine (IXCHIQ) has been licensed; however, post-licensure surveillance has identified serious adverse events following vaccination. In this study, the authors developed a dynamic transmission model to simulate chikungunya outbreaks across a range of transmission settings (from low to high) and vaccination strategies (traveler vaccination and outbreak response immunization) to evaluate the balance between vaccine benefits and the risks associated with adverse events. The authors found that the benefits of vaccination can outweigh the risks depending on the vaccination strategy, age group, and transmission intensity. These findings provide valuable evidence to inform policy decisions regarding the use of the vaccine, particularly given the recent safety concerns identified through post-licensure surveillance.

Overall, the study addresses an important and timely public health question. The modeling framework is well structured, clearly presented, and appears to fit the Brazilian data well. However, there are several aspects of the model that require further clarification.

Major comments

1. Initial conditions of the transmission model

It is unclear how the initial conditions of the dynamic transmission model were specified, particularly the number of initial infections used to seed the epidemic. What was the assumed number of initial infections, and did this value vary across states? **Additional details on how these initial conditions were chosen would improve the transparency and reproducibility of the model.**

2. Transmission intensity used in age-specific infection estimates

When calculating the number of new infections within each age group, it is not clear how the transmission intensity was determined. Is this quantity derived directly from the fitted transmission model? Alternatively, does it represent the cumulative force of infection over the 2022 Brazilian epidemic? **Clarifying the relationship between the transmission model and the age-specific infection calculations would strengthen the presentation.**

3. Estimation of the initial susceptible population

I had difficulty understanding how the initial susceptible proportion was derived (line 255).

Specifically, does the state-specific force of infection (λ_s) represent the cumulative force of infection over the outbreak?

More broadly, I have concerns regarding the use of a catalytic model to estimate the susceptible population. The catalytic model assumes that the force of infection is constant over both age and time, an assumption that is unlikely to hold for chikungunya. Unlike highly endemic infections, chikungunya transmission is typically characterized by intermittent outbreaks, resulting in age-specific seroprevalence profiles that are more likely to exhibit step-like patterns reflecting past epidemics rather than smooth age-dependent increases (e.g., <https://doi.org/10.1016/j.vaccine.2022.11.028>).

This distinction is particularly important in the context of this simulation study because pre-existing immunity strongly influences the age distribution of infections and, consequently, the estimated age-specific benefits and risks of vaccination. At a minimum, the authors should acknowledge this limitation in the discussion. Ideally, they should perform a sensitivity analysis using a more realistic step-like seroprevalence profile, with the magnitude of each step informed by the attack rate of previous outbreaks, to evaluate the robustness of their conclusions.

Minor comment

The manuscript contains a large number of figures, making it difficult to synthesize the main findings. I suggest adding a summary figure that integrates the key results and clearly highlights the combinations of transmission setting, vaccination strategy, and age group for which vaccination provides a net benefit after accounting for vaccine-associated adverse events. Such a figure would substantially improve the accessibility and policy relevance of the results.

Any attachments provided with reviews can be seen via the following link: